

Corporate Valuation 2409 - Transcription du cours

Transcription nettoyée - Cours magistral - Version brute sans structure - Le prof parle

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To help you understand the concepts, but if you will not work on your project it will become difficult to do the exercise in practice, yeah. Here we have the formulas, the formulas are very simplified versions, OK. We said we take the NOPAT or operating profits or we take the net income but when you're gonna get into the financial statements, there will be many items that you need to deal with that are not being shown in this simplified example, OK. So you can come across every... like... so please do that, OK. Find a company that you're gonna work on and start working on the company.

So by the end of today's session we should be able to prepare the financial statements for valuation and prepare the inputs for the valuation which includes forecasting as well as calculating the cost of capital which includes cost of debt and cost of equity, OK. So do that as soon as possible. Well for me there is no deadline for you, OK, it's up to you when you're ready, when you feel ready to start working on the project. My recommendation is that you should work in parallel, session by session, on the project. Don't wait for the last minute when you have to do everything from scratch, OK. So if you will work in parallel, one benefit would be that it would be the application of all the topics that we are working on in class and then the difficulty that you face, you will have the time to ask the questions, solve the problems and move on, OK.

So for the students who don't have the background in corporate finance or who have not taken the prerequisite courses, reach out to me. I have created what we call the corporate finance prerequisite that includes all the corporate finance topics, OK, many students have already done that. If someone needs to go back, refresh their concepts, calculating cost of capital, cost of equity, equity valuation, time value of money, that should be clear for everyone, OK.

OK so here we are going to continue with the first session, I didn't finish the first session so I will quickly wrap it up before the next one. So we talked about the reinvestments, OK, that's what we are gonna calculate, so here we stopped. So what are the reinvestments? This is the net CapEx. So net CapEx is the firm's investments into the physical assets, OK. So how do the firms make investments into the physical assets? How do you identify or do you extract this information from the financial statements? OK so this is the capital expenditures, investments in the firm's assets that have a life of more than one year and how do you find that information? From the cash flow statements, OK. What are the line items in the cash flow statements? But here my question is more specifically, you need to find the reinvestments that the companies make and how you are going to identify those reinvestments. For example, the baseline item that we use is property, plant and equipment, OK, property, plant and equipment, this is the fixed assets that the companies have. So is it the only one that you use? Why do we make the reinvestments? We make the reinvestments because we want to grow and in order to grow we have to reinvest, OK.

In order to grow, what are the ways through which the company can grow? So for example, purchase of equipment, opening new stores, geographic expansion, yeah. So I will call it more broadly, intangible, OK, I will come on that point in details afterwards, not for now. So the firms want to control, they want to open new stores, they want to expand geographically, how do companies do that? The company makes capital budgeting projects, the companies invest here, OK. But this is not the only way that companies grow. There is another way, not for all companies, but there are companies who also choose to grow through acquisitions, OK. So these are the choices that the companies make, either they grow organically, OK, we call it internal growth, they make capital budgeting projects in order to grow, but some of the companies

they choose to grow inorganically, we also call this external growth. So when we take the reinvestments, when we calculate the CapEx, we do not only look at property, plant and equipment or the changes in the property, plant and equipment, that would not be the best way to calculate the company's investments. This is only one way, both are not mutually exclusive, firms grow internally as well as externally, so we need to take into account both internal and external investments when you're gonna calculate the net CapEx, OK.

So the net CapEx plus the changes in the net operating working capital or changes in working capital. So the firms also need to invest into the working capital, the firms are gonna grow, they are gonna invest more, ultimately their working capital requirements also increase and we need to invest that. So you get the balances. So what is the working capital? I hope this is clear for everyone. So when you calculate the change in the working capital, you get two balances, negative working capital or positive working capital. What does negative working capital mean? Does the company need to fund its working capital? Current assets minus current liabilities, if you get negative working capital balance, what does that mean? It means your current liabilities are greater than your current assets. So what does that mean? You need to invest or there are two possibilities: either you need to invest in your working capital or your working capital would generate cash flow for you, OK. So negative working capital means your current liabilities are greater than your current assets, so your working capital is generating cash flows for you, OK, you do not need more investments and that will be adjusted in the formula, that would be adjusted in the formula when we are going to subtract the reinvestments from your operating profits. And positive working capital means you need to inject more cash, your current assets are greater than your current liabilities, so you need to fund that, you need to fund the working capital, which is normal, when the company grows, the more growth, the more working capital requirements, so the more financing you need.

So the net CapEx, this is adjusted for depreciation and amortization, OK, the total investments into fixed assets, part of since these assets we are going to use over the life of the assets, so every year we are going to depreciate or amortize, we are going to reduce the value of the assets by the amount for which we are going to utilize them. So the full cash flow formula, if you have to calculate the free cash flow for the company, the cash flows for both equity holders and debt holders, you have operating profits minus reinvestments. So what are reinvestments? That's equal to CapEx minus depreciation and amortization, so CapEx is adjusted for depreciation, and we add the changes in working capital, OK, so that would be the free cash flow to the company. If you have negative change in working capital, we said that's a source of cash flow for the company, so that would become positive and that would be added here, that would increase your cash flows, OK. So economically, if you don't need a formula, this is operating profits minus reinvestments, that would be the cash flows for the company and these cash flows will be used by equity holders and debt holders later on.

So here, if you have EBIT and tax rate, if it is 260 million, if tax rate is 24%, depreciation and amortization is 45 million, capital expenditure is 70 million, increase in working capital, this information would give you the sign about the working capital if it's positive or negative and you calculate NOPAT. NOPAT is you have operating profitability which is EBIT and you pay taxes, OK, operating profitability and you deduct taxes and you get net operating profit after taxes and you calculate reinvestment. So reinvestment here is capital expenditures, investments in your fixed assets, minus depreciation, you're gonna adjust depreciation and you're gonna add the change in working capital and you will be able to calculate free cash flow. This is simplified version just to understand formula but rest assured that for the company you're going to work on, it will not be that simple, so therefore it's important that you pick up a company and start working on it.

So two cash flows that we calculate: we have whether we are going to value the business or we are going to value equity, we are going to calculate free cash flows to the firm which is available for both equity holders as well as debt holders and if you're gonna value equity, then we can calculate free cash flows to equity, OK, so this cash flow is only for equity holders, so this cash flow we obtain after meeting claims of debt holders. So for free cash flow to equity, our starting point is operating profitability after taxes, we do

not pay debt holders, we do not pay interest expense since now we are calculating free cash flow for equity holders, so our starting point is...

OK, what is the objective here? Because it's just one time, it's just one time it could affect your valuation, here objective is that we want to get the EBIT, we want to get operating profitability, we need operating profitability which is sustainable, OK, which is sustainable over period of time. If there is a one-off item, if there is an exceptional item, we need to adjust.

The source of the negative profitability, OK, of course it's an indication the company is not doing good but they go in details, they check if they got negative EBIT because they have a huge restructuring charge in a given year, doesn't matter, OK, for them it's no problem, it's because of that exceptional one-off item, so they do not penalize the company because of negative profitability, because for valuation they don't look at company from accounting perspective. From accounting perspective, company is reporting negative profitability, perhaps company is not doing good, but no, they do not worry about it because they use normalized EBIT, so they use EBIT that's gonna be generated or that can be sustained in future, this negative image is just one-time off and it's gonna go back to normal soon, this is approach we use here. We want an EBIT that could be sustainable, OK, whether it's negative or positive doesn't matter, so you want to get sustainable operating profit. So in this case, EBIT reported is 510, do we need to make an adjustment? Yes, this is an adjustment, it's a one-off restructuring charge, so 30 million that could be added back, so normalized EBIT would be 540, something like that, OK.

Whatever assumptions that you make, they should not be any ad hoc assumption, it should not be assuming that it's going to grow at 5%, whatever it is, so there should be some evidence that should be backed by your assumption, that assumption should be defensible, OK, so it's coming from management statements, earnings calls, or investors, or on financial statements they provide information, management discussion and analysis they provide information, or company websites, they have their presentations where they provide their investment plans, OK.

So normalized EBIT, we are back here, we only see simplified case, we only see one charge, calculate NOPAT, net operating profit after taxes, so once we have normalized earnings and then we just apply taxes, since these are cash flows for company, so normalized EBIT times 1 minus tax rate, so if it is 540, what is tax rate? 1 minus 0.25, so you will get NOPAT, after paying taxes you will get operating profitability. Now you have to calculate free cash flows to the firm, not for equity holders, free cash flows to firm, but starting point we start with NOPAT, OK, so NOPAT minus reinvestments. So what is reinvestment? 60,000, net CapEx plus change in working capital, OK, so what is net CapEx? This is total capital investments adjusted for depreciation and amortization, OK, so do they provide that information? So you have CapEx and you have adjustment for depreciation, so CapEx is 105 minus 65, 65 did you get net CapEx and you add increase in working capital which is 20, 60, OK, you got 60, right? Yeah, yeah, so we subtract reinvestment and we get free cash flow to firm, 345. So what is reinvestment rate? From this information that you have on slides, reinvestment divided by NOPAT, so what is reinvestment? You invest part of profitability in new projects, so that would be CapEx plus change in working capital divided by NOPAT. So reinvestment rate, say 60 divided by NOPAT 405, so it's 14.81%, OK. So what it means is company is reinvesting 14 cents per dollar of operating profitability, OK, company is reinvesting.

Return on invested capital, ROIC, NOPAT divided by invested capital. So what is invested capital? Is it total assets or what? What do you do? So you need to calculate invested capital, beginning invested capital, OK, and then you get ROIC. So this is return that company generates on its investments. And then growth, fundamental growth, g equals ROIC times reinvestment rate. So if ROIC is high and you reinvest more, you grow more, OK. So this is sustainable growth.

For your company that you're gonna work on, you're gonna calculate, you're gonna find segments that you will have, you're gonna calculate sales for each segment, OK, how much each segment is going to contribute and how it's going to grow, but for everything you should have evidence, OK. So if you have

sales for each segment, it's based on historical data, so you will have evidence for that because you have historical, if it's coming from increase in market size, market shares, there is new investment project, they're gonna open new stores, they're gonna launch new product, companies, managers do provide guidance, many sources now available, more specifically if you're working on US companies, so much information is being generated by analysts and by industry as well, OK. Well here objective is that you're gonna make forecast and you need to bring it towards maturity, when you're gonna calculate terminal value of company, terminal value of projects, so that should be moving, all forecasts should be moving toward maturity, OK. You should have this table, even if it's a very high growth company, at some point in time you should be able to calculate sustainable growth rate that's gonna be sustained in future without having very precise information. Second component that we calculate is operating margins, to forecast operating margins we look at different factors, pricing mix is one of main factors that contribute towards how operating margins are gonna evolve over period of time or in future. So here question is, OK, if we are even if we are not introducing new products, there is no new investment project that companies have, it's a mature company, so we look at therefore understanding business model is important.

It's a capex that becomes part of cash, R&D is also treated, yes, I think there are two ways to treat it, one you can treat some aspects as operating expense, for example salaries, OK, let's keep it simple, how do we do in accounting? Irrespective of what it is, if we expense it now, OK, so this is what we do in accounting, so you have sales, you have expenses and R&D is an expense, we expense it, this is from accounting perspective. From valuation perspective, what should we do? Should we expense it? Do you think so? Again going back to basic accounting principle, there is matching principle, OK, so we have revenues and we match expenses with revenues of that accounting period. If you invest 100 million in R&D, is there any matching between revenues and expenses? So you have an expense for a given year, let's say 100 million and then you have sales for 2000 million dollars and you subtract it, so do you match that expense with same time? Sorry, you need to treat it differently from accounting. In accounting we expense it out, OK, so then couple of slides, so this is one of important adjustments that we make when you are gonna restate statements, not for all companies, more specifically if you're going to select company from tech sector, from pharmaceutical sectors and some manufacturing companies, they invest into R&D, OK, from accounting point of view we expense it out, but it doesn't work for valuation because R&D is an investment that we've made, we are going to have new effect, we are going to have new products or new features in existing products and that's going to generate revenue over period of time, so we need to capitalize R&D. R&D investment does not show up in your balance sheet because it goes in income statement as expense and then goes to cash flow statement and that's it. So what we do is we capitalize R&D, we create an asset, we amortize over useful life, OK, so that will increase EBIT, increase invested capital, etc.

Investments, you have side, at some point in time return on invested capital is high while it's lower on other side, so then perhaps we take average and then we start making prediction, make growth and reinvestment consistent. So growth and reinvestments should be consistent, so companies want to grow, you make forecast and then you show that company is going to grow in future, your revenues are going to grow, your operating margins are going to grow, again so then your reinvestment should also grow. If you forecast sales are growing, margins are growing but your investment does not grow, so then you are not going to get that desired results, there is inconsistency, so you don't have very good story for that, so growth is not free, you need to invest, OK, without reinvestments you will not have high return on invested capital, that also tells you how much investment they need. If you are efficient in using resources, efficiency being captured by return on invested capital, so then you need lower reinvestment, OK, with lower reinvestments you are able to generate more cash, more profit for companies. If you are inefficient, it's not only because we have bad managers, it also depends on market, depends on business, depends on lifecycle of business, when business is at mature stage, their margins increase, their returns increase, so then they need to invest more, then they generate more cash flows and they invest more and they grow. So high ROIC requires lower reinvestment because with lower investments you are able to generate

higher growth, you are able to grow more. If you have low ROIC, you are not using your investments efficiently, you need more reinvestment to grow.

The company's current capital structure and target capital structure are fundamentally different, that is generally the case for growth companies, growth companies they tend to have more investment opportunities, they are more risky, so cost of capital may be very high today but they're moving towards optimum capital structure, maybe financing those companies with more debt is expensive still but currently required rate of return is high now, so we make adjustment for that, OK. So now, of course, it's valuation based on assumption, whatever you're gonna do, even when you're gonna forecast revenues, revenues will be based on assumptions, everything that you do there is uncertainty, there is risk of whatever you're doing, so we do not only have baseline case, OK, we do not hide all uncertainty in baseline case, we then start stressing out our assumptions, we print scenarios, OK, what's gonna happen in best case, what's gonna happen in worst case, OK, so this is normal case and this is normal scenario, this is value that you get if you go base and best case, so what would be value of company, you get ranges, OK, so these are steps that we follow when we prepare valuation. So our forecast again, whenever you're gonna make forecast of financial statements, we do not make forecast for each line item independently, again as I said previously, you are gonna tell economic story, you do valuation and you are gonna tell economic story that should be correct, there should be consistency, whatever assumptions that you make, you say OK, your revenues are going to grow, for example, this is story that you tell, your analysis says revenues are going to grow but you don't see growth in reinvestments, then inconsistency, OK.

The determination of bonds today, so at what price you will be able to issue bonds, OK, so you have coupon rate and then you have yield to maturity, so benchmark rate is yield to maturity that you have to date or credit spread, so that could be also way to calculate. So if your company, we look at probability of default, OK, if your company does have credit rating, then that allows you to calculate cost of debt, right, and we also need tax rate because there are benefits from debt financing which is tax shield. The weights that we use, so there are two different types, so weights are based on value of equity and value of debt, OK, so there are two values that would be available for you, both for equity and debt, one is book value, another is market value, value of equity in your financial statements, in balance sheet, value of debt in your balance sheet and market value, market capitalization and valuation of debt, OK, value of your bonds today, trading price of your bonds today. So here you have component, this is weight of equity and this is weight of debt in total financing of company, what values do you take? Do you take book value or market value? After calculating cost of equity you also need weight, OK, so you have two values, one is book value, this is value that is being recorded in your financial statements, open financial statement balance sheet you will find common equity, you will find numbers, you will find debt, you will find total value that's being recorded in your financial statement, this is actual price that company paid, OK, when it issued bonds, and then you have market value, price at which shares are being traded in market. So what value do we take here? Market value, well book value looks more realistic, OK, so this is price at which we issued debt, OK, but we are trying, but at that time when we issued debt, for example, we take debt value, so this is what is being recorded, we have like 5 years before, so five years before riskiness of your company was very different as of today, OK, so now value of debt in your books, we don't take book value neither for debt nor for equity, from financial valuation point of view it's irrelevant now because company has evolved, perhaps it's not same, what we need to do is we need to calculate value today, so we have market value, if your bonds are being traded in capital markets, you have value for bonds, so we take market value, we always take market value to calculate weight of each component of financing, never book value, OK, although book values are more stable.

Financing, so it's based on target capital structure, now what if we cannot calculate plan for company with current capital structure, OK, so what to do? I cannot use current capital structure of my company because my current capital structure is different from target capital structure and what to do? Market value, sorry, market value, yeah, like look at peers, maybe that could be another option and it happens, it happens often, we cannot use our current capital structure, we cannot use our current weights because we are

gonna go into several changes, OK, so current capital structure, current cost of capital is not reflection of what would be cost of capital going forward, it's gonna be involved over period of time, then whatever changes that you foresee, you could look at your peers, OK, that have similar infrastructure and so on and similar changes, you're gonna identify comparable company or similar company and you calculate each input for those companies, OK, I'm gonna talk about that, let me finish with this. So here we said country risk, now country risk depends on what, OK, we adjust cost of equity for exposure to foreign markets, exposures to emerging economies are riskier, so there's additional risk for investors which is coming from specific countries and we adjust for that, increase cost of equity by amount of additional risk that they take. So this risk is not... in practice, OK, otherwise we take if it's US companies we take only... but it shouldn't be based on where company is headquartered, it should be based on firm's exposure to foreign countries. If firm's exposure is US, firm's exposure is European markets, OK, there is no country risk, won't have same level of country risk, same level of country characteristics, so there is no additional exposure, there is no additional risk that is specific to European countries, but if US company has exposure to emerging markets, China, Brazil or other less developed markets, there is additional risk that we take, so country risk is not normally specific to where company's headquarters is, for example if you are working on your company and if you are calculating cost of equity, so you're gonna look at current... if they have subsidiaries in emerging markets, so US company is exposed to emerging market, we need to make adjustment. So we use default free rate in same currency as cash flows, but if it's euro cash flow we use euro risk free benchmark, if it's US we use US and we do not use local government yield only because company is headquartered, OK, so all components that you're gonna...

OK, so that's it for today, please start working on your project, yeah.